

Data structure

- list - tuple # more than 1 variable - set - dictionary - range

list []

```
In [1]: l=[]  
l
```

```
Out[1]: []
```

```
In [2]: len(l)
```

```
Out[2]: 0
```

```
In [11]: l1= [10,20,30,40,50]  
l1
```

```
Out[11]: [10, 20, 30, 40, 50]
```

```
In [12]: len(l1)
```

```
Out[12]: 5
```

append

```
In [13]: l1
```

```
Out[13]: [10, 20, 30, 40, 50]
```

```
In [14]: l1.append(60)  
l1.append(70)  
l1
```

```
Out[14]: [10, 20, 30, 40, 50, 60, 70]
```

```
In [15]: l1
```

```
Out[15]: [10, 20, 30, 40, 50, 60, 70]
```

```
In [16]: len(l1)
```

```
Out[16]: 7
```

```
In [17]: l2=[2,3.3,'pragati',True,(3+4j)]  
l2
```

```
Out[17]: [2, 3.3, 'pragati', True, (3+4j)]
```

```
In [18]: len(l2)
```

```
Out[18]: 5
```

```
In [19]: print(l)
         print(l1)
         print(l2)
```

```
[]
[10, 20, 30, 40, 50, 60, 70]
[2, 3.3, 'pragati', True, (3+4j)]
```

```
In [20]: print(len(l))
         print(len(l1))
         print(len(l2))
```

```
0
7
5
```

```
In [21]: print(id(l))
         print(id(l1))      # in different id
         print(id(l2))
```

```
2195767163328
2195773258112
2195767344768
```

```
In [ ]: # list inside list means nested list
```

```
In [27]: l1 = [10, 20, 30, 40, 50, 60, 70]
         l1
```

```
Out[27]: [10, 20, 30, 40, 50, 60, 70]
```

```
In [28]: l1.append(['a', 'b', 'c'])
         l1      # double run then added ['a', 'b', 'c'] one more
```

```
Out[28]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]
```

```
In [29]: len(l1)
```

```
Out[29]: 8
```

```
In [30]: print(type(l1))
```

```
<class 'list'>
```

list membership - in - not in

```
In [31]: l3=[2,4,5,7,6]
         l3
```

```
Out[31]: [2, 4, 5, 7, 6]
```

```
In [32]: 2 in l3
```

```
Out[32]: True
```

```
In [33]: 8 in l3
```

```
Out[33]: False
```

```
In [34]: 4 not in l3
```

Out[34]: False

In [35]: 13

Out[35]: [2, 4, 5, 7, 6]

In [37]: 14=13.copy()
14

Out[37]: [2, 4, 5, 7, 6]

In [38]: print(id(13)) # if same number ,in datastructure id are different
print(id(14)) # if same number ,in datatype id are same

2195773553408

2195773268288

In [39]: 14

Out[39]: [2, 4, 5, 7, 6]

In [40]: len(14)

Out[40]: 5

In [42]: del (14) # permanantly deleted
 # if i double run on it,it gives error bcoz list has already delete

In [43]: 11

Out[43]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]

In [44]: 11.append(10,400) # it gives error bcoz, we csn pass only 1 argument at a

```
-----
TypeError                                 Traceback (most recent call last)
Cell In[44], line 1
----> 1 11.append(10,400)                    # it gives error bcoz, we csn pass only 1 argumen
t at a time

TypeError: list.append() takes exactly one argument (2 given)
```

In [45]: 11

Out[45]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]

In [47]: 11.append(10)
11.append(30) # i run it 2 times that's why 10 & 30 is repeated
11

Out[47]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c'], 10, 30, 10, 30]

In [48]: 11.count(10)

Out[48]: 3

In [49]: 11.count(20)

Out[49]: 1

```
In [50]: l1=[10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]
l1
```

Out[50]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]

indexing

```
In [51]: l1
```

Out[51]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]

```
In [52]: l1[2:6]
```

Out[52]: [30, 40, 50, 60]

```
In [53]: l1[3:8]
```

Out[53]: [40, 50, 60, 70, ['a', 'b', 'c']]

```
In [54]: l1[2:60]
```

Out[54]: [30, 40, 50, 60, 70, ['a', 'b', 'c']]

```
In [55]: l1[1000]
```

```
-----
IndexError                                Traceback (most recent call last)
Cell In[55], line 1
----> 1 l1[1000]

IndexError: list index out of range
```

```
In [56]: l1[3:1000]
```

Out[56]: [40, 50, 60, 70, ['a', 'b', 'c']]

```
In [57]: l1
```

Out[57]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]

```
In [58]: l1[-6:]
```

Out[58]: [30, 40, 50, 60, 70, ['a', 'b', 'c']]

```
In [60]: l1[3:-2]    # count from starting & ending, remaining value
```

Out[60]: [40, 50, 60]

```
In [61]: l1
```

Out[61]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]

```
In [62]: l1[0:6:2]
```

Out[62]: [10, 30, 50]

```
In [64]: l1[ : : 4]      # start to end whole consider
```

```
Out[64]: [10, 50]
```

```
In [67]: l1[ : : 1]      # whole
```

```
Out[67]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]
```

reverse

```
In [68]: l1
```

```
Out[68]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]
```

```
In [69]: l1[ : : -1]      # all reverse
```

```
Out[69]: [['a', 'b', 'c'], 70, 60, 50, 40, 30, 20, 10]
```

```
In [74]: l1
```

```
Out[74]: [10, 20, 30, 40, 50, 60, 70, ['a', 'b', 'c']]
```

```
In [81]: l1[-8: :3]
```

```
Out[81]: [10, 40, 70]
```

```
In [ ]:
```